BYZANTINE-RESILIENT FEDERATED LEARNING FOR TACTICAL UAV SWARMS OVER LOSSY WIRELESS LINKS

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in Electronics and Communication Engineering

by

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APPROVAL OF THE VIVA-VOCE BOARD

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This is to certify that the project report entitled **Byzantine-Resilient Federated Learning for Tactical UAV Swarms over Lossy Wireless Links**, submitted by **Nasika Sasank** (23ECB0A03), **Aaron George** (23ECB0F06), **Sachin Sreekumar Nair** (23ECB0F11) to the National Institute of Technology Warangal, is a record of bona fide work carried out under my supervision in the Department of Electronics and Communication Engineering during the academic year 2026–2027, and I consider it worthy of consideration for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering of the Institute.

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DECLARATION

We certify that

a. The work contained in this report is original and has been done by us under the general supervision of our supervisor.

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in writing the report.

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Abstract

Federated learning (FL) allows unmanned aerial vehicles (UAVs) to train a shared image classifier by sending model updates to ground stations. Two failures can occur in the same round: wireless packet loss and deliberate poisoning by Byzantine clients. The base paper studies erasures with honest clients [1]. We study erasures and poisoning together.

The PyTorch simulator distributes aerial-transformed CIFAR-10 images among ten UAVs with different class proportions. Three base stations aggregate locally and synchronize every five rounds. The channel replays relative power from DeepSense 6G Scenario 23 around assumed signal-to-noise ratio anchors with virtual neighbouring stations. In each attack setting, three UAVs use the same poisoning strategy (sign flip, ALIE-style perturbation or inner-product manipulation). Our aggregator, CCRA, scores updates against a trusted reference. It masks lost coordinates and uses link telemetry to slow trust recovery on weak links.

FedAvg reached 62.6% with IID data and 60.2% with non-IID data. After a single-seed pilot, we separated masking and telemetry in a prespecified 720-run study spanning six aggregators, four attack settings, three channels and 10 paired seeds. The primary telemetry stress interaction was -2.7 percentage points. Its 95% paired-seed bootstrap interval $[-3.9, -1.5]$ lay below the prespecified -1 point threshold. Under attack at 5 dB, masking added 4.2 points over the zero-filled baseline ([2.2, 5.9]). Adding telemetry changed accuracy by -3.5 points ($[-5.1, -2.1]$) and increased honest-update rejection from 13.8% to 41.2%. The constant-evidence drift calculation shows how asymmetric rewards and penalties can push equilibrium trust toward the lower tail of honest scores. The four methods using a trusted reference had zero divergences. FedAvg and the median had 41 and 26, respectively, in 120 runs each.

In the separate 240-run follow-up, the base paper's fresh-only and reuse-old rules diverged in 41 and 42 of 120 runs, respectively, including every sign-flip run. Fresh-only led reuse-old by 5.4 points on clean 5 dB runs. At 5 dB under attack, mask-only CCRA and FLTrust led fresh-only by 19.2 points ([15.9, 23.0]) and 19.8 points, respectively. Without attack, the signed mask-only gap was -2.1 points ([-2.9, -1.4]). These contrasts combine differences in aggregation design and execution environment. The follow-up ran on Windows GPU workers. In the ideal-channel cross-check, the runs agreed within 1.1 points.

The results support masking under severe packet loss and identify a drawback of telemetry-weighted rewards. An IEEE letter manuscript presents these findings. Phase II will revise the reputation rule, test it with fresh seeds, higher-mobility traces and adaptive adversaries, and repeat the follow-up comparisons on shared hardware.

Index Terms : Federated learning, UAV swarm, Byzantine robustness, packet erasure, hierarchical aggregation, reputation.

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List of Abbreviations

ALIE	A Little Is Enough (model-poisoning attack)
BS	Base station
CCRA	Context-Calibrated Robust Aggregation (the aggregator studied here)
CI	Confidence interval
CNN	Convolutional neural network
CosAvg	Cosine-similarity weighted averaging
EMA	Exponential moving average
EW	Electronic warfare
FedAvg	Federated averaging
FL	Federated learning
HFL	Hierarchical federated learning
IID	Independent and identically distributed
IPM	Inner-product manipulation (model-poisoning attack)
PDR	Packet delivery ratio
PER	Packet error rate
pp	Percentage points
RMS	Root mean square
SGD	Stochastic gradient descent
SINR	Signal-to-interference-plus-noise ratio
SNR	Signal-to-noise ratio
TEE	Trusted execution environment
UAV	Unmanned aerial vehicle

List of Symbols

K	Number of UAVs (clients)
B	Number of base stations
E	Edge rounds between global synchronizations
P	Number of model parameters
L	Parameters per packet
f	Number of Byzantine UAVs
$\mathcal{D}_k$, n_k	Local dataset of UAV k and its size
$\mathcal{D}_0$	Trusted root dataset held by each base station
$\mathbf{w}^t$, $\mathbf{w}_b^t$	Global model and edge model of BS b in round t
$\boldsymbol{\Delta}_k^t$	Model update sent by UAV k in round t
$\mathbf{m}_k$	Delivery mask: 1 where the packet carrying a coordinate arrived
$\mathbf{r}$	Reference update computed on $\mathcal{D}_0$
γ_k, $\bar{\gamma}$	Uplink SNR of UAV k; reference SNR of the trace
p_k	Packet error probability of UAV k
a, γ_{th}	Slope and threshold of the packet-error curve
s_k	Norm-clipping factor
c_k	Masked cosine similarity to the reference update
b_k	Behaviour score in $[0, 1]$
T_k	Reputation (trust) of UAV k
e_k	Link evidence in $[0, 1]$ that slows reputation rewards
β, τ	Reputation memory and exclusion gate
ω_k	Aggregation weight of UAV k
α	Dirichlet concentration of the non-IID data split
λ, z, ϵ	Strengths of the sign-flip, ALIE and IPM attacks

Chapter 1

Introduction

1.1 Background

Small unmanned aerial vehicles (UAVs) are now used in groups, or swarms, for surveillance, search and rescue, disaster assessment and border patrol. Each UAV carries a camera and collects images of the area it covers. Training across the datasets of several UAVs may improve generalization compared with training on a single UAV's data [2], but sending raw imagery to a ground station is costly in bandwidth and exposes sensitive data if a link is intercepted.

Federated learning (FL) addresses this by moving training to the data. Each UAV trains a copy of the model on its own images and uploads only the resulting model update. A server (here, a ground base station) combines the updates into a new global model and sends it back. The raw images remain on the UAV, and an update is usually much smaller than the data it was computed from. Updates can still leak some information about the data, so FL alone is not a privacy guarantee.

A tactical swarm, however, breaks two assumptions that standard FL relies on.

1. **Links are unreliable.** Air-to-ground channels fade as the UAV moves, hand over between base stations, and may be deliberately jammed. Parts of an update are lost, and some updates miss the round deadline entirely.

2. **Some clients may be hostile.** A captured or spoofed UAV can follow the protocol while sending a poisoned update designed to damage the global model. Such clients

are called *Byzantine*.

Most Byzantine-robust aggregation rules assume that every update arrives intact, and most work on FL over imperfect channels assumes that every client is honest. A swarm needs both at once.

1.2 Problem Statement

Given a UAV swarm that trains a shared image classifier with hierarchical FL over lossy air-to-ground links, in which a minority of UAVs are Byzantine, design and evaluate an aggregation rule that (i) keeps the global model accurate and stable, (ii) does not penalize honest UAVs simply because their links are poor, and (iii) can be compared fairly against established rules under identical conditions.

1.3 Aim and Scope

The aim of this project is to determine *which* kind of channel awareness actually helps a robust aggregator in a UAV swarm, and to quantify the effect with a controlled, reproducible experiment. Its scope in Phase I is:

- a simulation of hierarchical FL with up to three base stations, UAV association and handover;
- packet-level uplink losses from a synthetic jammed channel and from a measured UAV received-power trace;
- five supported attacks (label flip, sign flip, Gaussian noise, ALIE and IPM), of which the preliminary study uses label flip and the controlled study the three model-poisoning attacks sign flip, ALIE and IPM;
- six aggregation rules evaluated on the same data, models, attackers and channel realizations, and the base paper's two erasure rules for a separate comparison.

Hardware flight tests, real radio transmission and attacks that forge link telemetry are outside the Phase I scope.

1.4 Objectives

1. Build an FL baseline for a UAV swarm on aerial-transformed CIFAR-10 and measure its accuracy under IID and non-IID data.
2. Model a lossy uplink in which lost packets erase parts of an update, and add Byzantine attackers.
3. Design a reputation-based aggregator (CCRA) that uses a trusted reference update, and give it two forms of channel awareness: erasure masking and telemetry-weighted reputation.
4. Separate the effect of each component with a prespecified, multi-seed ablation study, and report the results with honest uncertainty estimates.

1.5 Contributions

- An open, tested PyTorch framework for hierarchical UAV-swarm FL with trace-driven packet erasures, handover and five attacks, with automated tests.
- A preliminary study, then a controlled 720-run study whose analysis plan and decision threshold were fixed before the runs.
- The finding that erasure masking improves accuracy under attack on weak links by 4.2 percentage points, while telemetry-weighted reputation lowers it, together with an analysis of the reputation rule that suggests a mechanism consistent with this result.
- A completed 240-run comparison of the base paper's two erasure rules on matched seeds, attackers and channels, with validated checkpoint recovery and an explicit separation from the prespecified ablation.
- An IEEE letter manuscript presenting the ablation and the erasure-rule follow-up.

1.6 Organization of the Report

Chapter 2 examines how prior studies handle poisoned updates and unreliable uplinks. Chapter 3 defines the simulator and aggregation rules and derives the effect of link evidence on the reputation update. Chapter 4 traces the experiments from the FedAvg baseline to the controlled ablation and the erasure-rule follow-up. Chapter 5 draws on those results to set the Phase II priorities. Appendices A and B give the simulation settings and reproduction commands.

Chapter 2

Literature Review

2.1 Federated Learning

FedAvg provides the baseline for comparing aggregation rules in this project. In the original algorithm, participating devices start from the same model and run local stochastic gradient descent. The server combines their resulting parameters in proportion to the number of examples at each device [2].

A sample-count weight says how much data a device holds. It says little about which classes the device sees. Hsu et al. [3] varied class imbalance with Dirichlet sampling and observed lower FedAvg accuracy on CIFAR-10 as client label distributions became more dissimilar. In our simulator, each class is distributed across the UAVs using a Dirichlet draw with $\alpha = 0.5$.

Abad et al. [4] separate aggregation within each cell from network-wide synchronization. Their small-cell base stations combine updates from their own devices and periodically coordinate through the macro base station. In this project, ground base stations serve as the cell aggregators, with one swarm-wide synchronization for every five local rounds in the controlled study.

2.2 Byzantine-Robust Aggregation

A single malicious client can move the FedAvg average arbitrarily far, so a large body of work replaces the mean by a robust statistic.

- **Krum** [5] selects the update whose summed distance to its nearest neighbours is smallest. It requires $n \geq 2f + 3$ clients to tolerate f attackers.
- **Coordinate-wise median and trimmed mean** [6] take the median, or a trimmed mean, of each parameter separately, and come with statistical error guarantees.
- **FLTrust** [7] gives the server a small clean root dataset. The server computes its own update on that data each round, scores each client by the (clipped at zero) cosine similarity to it, and rescales every client update to the norm of the server update before averaging.
- **Geometric median** (RFA) [8] aggregates with an approximate geometric median, which differs from the coordinate-wise median.

Non-IID data make honest updates look different from each other, which a robust rule can mistake for an attack; bucketing [9] adapts robust rules to heterogeneous data. Attacks that accumulate small changes over rounds are countered by using history, through worker momentum and iterative clipping [10]; this is related to, but different from, keeping a reputation per client. These rules are evaluated against attacks of increasing strength. A sign-flip attack sends a scaled negative of an honest update. ALIE ("A Little Is Enough") [11] shifts each coordinate by a small multiple of the honest standard deviation, which is enough to evade distance-based rules. Inner-product manipulation (IPM) [12] sends a negative multiple of the honest mean so that the aggregate's inner product with the true gradient becomes negative.

Because the same UAVs participate across rounds in our simulator, a base station can use their earlier updates when assigning trust. This history also gives an adaptive adversary a way to manage its reputation. Liu et al. [13] formulate the decision to send a benign or malicious update as a partially observable Markov decision process. In their experiments, an attacker able to vary its behaviour across rounds evaded reputation-based defences and reduced model accuracy. The controlled study in this report uses fixed attack settings.

2.3 Aggregation with Wireless Erasures

Shirvanimoghaddam et al. [1], the authors of our base paper, ask what the server should aggregate when a device's upload is missing. They use the finite-blocklength normal approximation to set the probability of losing the entire upload. The server can omit that device for the round or fill in its most recently received parameters. The authors prove convergence to the error-free solution for reuse-old under conditions on the loss functions, step size and erasure probability. Their uniform-data example favours the rule that uses only fresh parameters.

Our base-paper comparison adapts both rules to uploads split across several packets (Section 3.4.1). The server can then receive only part of a model update. We also include Byzantine clients, whereas the original analysis considers honest clients. Barkatsa et al. [14] consider malicious clients that jam the non-orthogonal multiple access (NOMA) uplink of legitimate ones, and let every client choose its transmit power in a Bayesian game with incomplete information about the other clients' types. Li et al. [15] combine communication efficiency (1-bit compressed sensing of sign-based gradients), differential privacy and Byzantine robustness, but do not model lost packets. Byzantine robustness has also been studied for analog over-the-air aggregation, in which clients transmit simultaneously and the channel itself computes a sum: Sifaou and Li [16] aggregate robustly across random groups, Yao et al. [17] combine Byzantine identification, adaptive clustering and aggregation weights, and Fan et al. [18] use UAVs to collect gradients over the air from heterogeneous workers under Byzantine attack. In those systems the server cannot observe individual updates, whereas our UAVs send individually observable digital updates that may lose packets.

2.4 Federated Learning for UAV Swarms

Xiao et al. [19] propose fully decentralized FL for on-board missions of a UAV swarm: UAVs exchange models with neighbours over a time-varying graph instead of through a server, using an inexact stochastic parallel random-walk ADMM, and the framework is demonstrated on beamforming design. It considers neither adversarial UAVs nor packet

loss. Liao et al. [20] propose UAV-assisted FL in which the UAV aggregates inside a trusted execution environment (TEE), using a robust rule they call CosAvg: each client is scored by the sum of its pairwise cosine similarities with the others, and the top-ranked updates are averaged. Our preliminary study used a simpler CosAvg-style baseline that weights each update by its cosine similarity to the mean update; it is not an exact reproduction. Realistic UAV link data have recently become available: the DeepSense 6G dataset [21] includes a scenario with a flying UAV and a ground base station [22], which we use to drive packet losses in simulation.

2.5 Research Gap

Table 2.1 summarizes the reviewed work; it is a small sample and does not establish that no other work covers these points. Prior work addresses Byzantine robustness in conventional FL, where robust rules were defined for complete update vectors, and in analog over-the-air aggregation, including UAV-assisted architectures. Work on erroneous links, such as the base paper, assumes honest clients. This project studies a different setting: digital packet erasures in hierarchical FL with UAV clients, together with model-poisoning attackers. It is also natural to assume that link telemetry (signal quality, delivery ratio, handover state) should make a robust aggregator *fairer* to honest UAVs on poor links. This project tests that assumption with a paired ablation that separates delivered-coordinate masking from telemetry-weighted reputation.

Table 2.1 Coverage of the problem in the reviewed work. "Poisoned updates" means model- or data-poisoning clients; "digital erasures" means individually received updates that can be lost in whole or in part.

Work	Poisoned updates	Digital erasures	UAV setting	Measured channel
FedAvg [2]	–	–	–	–
Krum, median [5], [6]	✓	–	–	–
FLTrust [7]	✓	–	–	–
Erroneous links [1]	–	✓	–	–
Jamming mitigation [14]	–[a]	✓	–	–
DP + Byzantine [15]	✓	–	–	–
Over-the-air robust FL [16], [17]	✓	–[b]	–	–
UAV-assisted over-the-air FL [18]	✓	–[b]	✓	–
UAV-swarm FL [19]	–	–	✓	–
TEE + CosAvg [20]	✓	–	✓	–
This project	✓	✓	✓	✓[c]

[a]Malicious clients jam the uplink; they do not send poisoned updates. [b]Analog channel impairment of a superposed sum, not digital erasures. [c]Relative power variation of one measured link, with virtual neighbouring BSs and assumed SNR anchors.

Chapter 3

Theoretical Study

This chapter describes the simulated system, the attacker and the aggregation rules compared, and analyses the reputation update. Chapter 4 describes how they were evaluated. All of it is implemented in the project repository; file names are given where they help locate the code.

3.1 Hierarchical Federated Learning for a UAV Swarm

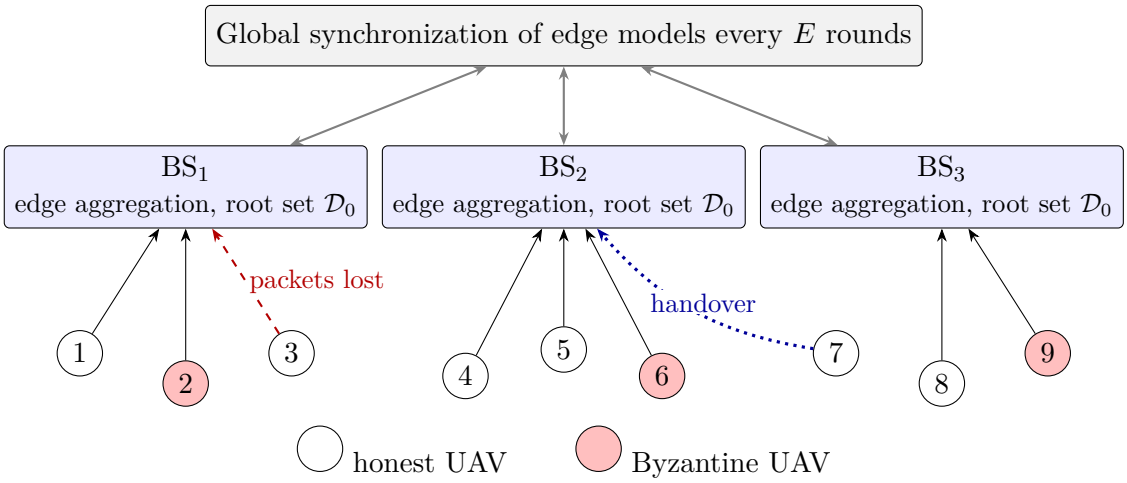


Figure 3.1 Hierarchical FL over a UAV swarm. Each base station aggregates the updates of the UAVs it serves, some of which arrive with lost packets; edge models are synchronized every E rounds.

Figure 3.1 shows the system. K UAVs are served by B ground base stations (BSs). Each UAV k holds a private dataset $\mathcal{D}_k$ of n_k images. In every round t:

1. each UAV attached to BS b starts from that BS's edge model $\mathbf{w}_b^t$ and trains it for

one local epoch of SGD on $\mathcal{D}_k$;

2. it uploads the update $\boldsymbol{\Delta}_k^t$ (new local model minus the edge model), which is split into packets of L parameters;

3. the BS aggregates the updates it receives and sets $\mathbf{w}_b^{t+1} = \mathbf{w}_b^t + \mathrm{Agg}(\cdot)$;

4. every E rounds the edge models are averaged, weighted by the number of training samples held by the UAVs in each cell, and the result is sent back to every BS.

UAVs attach to the BS with the strongest received power. A UAV hands over only when another BS is stronger by a hysteresis margin of 3 dB for two consecutive rounds, which avoids ping-pong handovers (`fl/topology.py`).

3.1.1 Dataset and Model

Images come from CIFAR-10, transformed to imitate some properties of aerial imagery (`fl/data.py`); we call the result aerial-transformed CIFAR-10, since the transformations do not make it genuine UAV imagery. Each image is rotated by a random multiple of 90°, since a downward-looking camera has no preferred orientation, and with probability 0.3 or 0.2 is down-sampled to 24×24 or 16×16 pixels and up-sampled again to imitate higher flight altitudes. Training images are split among the UAVs either uniformly (IID) or with Dirichlet(α) class proportions. In the controlled study of Chapter 4, 500 images (50 per class) are held out as the root set $\mathcal{D}_0$, which is assumed clean and never given to any UAV, and the remaining 49,500 are split among the UAVs. The earlier baseline experiments split all 50,000 images among the UAVs and used no root set.

The classifier, TacticalEdgeCNN, is a small convolutional network with three convolutional layers (16, 32 and 64 channels) and two fully connected layers, 156,074 trainable parameters in total and no normalization layers, chosen so that it could plausibly be trained on board. Without normalization layers the whole model state consists of plain weights, which keeps averaging across UAVs straightforward.

3.2 Wireless Uplink Models

The channel decides, for every packet of every update, whether it arrives (`fl/channel.py`). Two models are used.

3.2.1 Synthetic Jammed Channel

UAVs move by a random walk at 100 m altitude over an 800×800 m area. The received power from UAV k at its serving BS is

$$P_{\mathrm{rx},k} = P_{\mathrm{tx}} - \mathrm{PL}(d_k) + 10\log_{10}|h_k|^2, \qquad \mathrm{PL}(d) = 40 + 10\eta\log_{10} d \ \mathrm{dB}, \tag{3.1}$$

with $P_{\mathrm{tx}} = 20$ dBm, path-loss exponent $\eta = 2.2$, and Rayleigh fading $|h_k|^2 \sim \mathrm{Exp}(1)$. A ground jammer of power 10 dBm targets a sector of the swarm, and the SINR is

$$\gamma_k = P_{\mathrm{rx},k} - 10\log_{10}\big(N_0 + J_k\big), \tag{3.2}$$

where N_0 (-90 dBm) and the received jamming power J_k (zero for UAVs outside the jammed sector) are added as linear powers in mW before conversion to dBm. This jammed channel is used only in the preliminary study; the controlled study of Chapter 4 has no jammer.

3.2.2 Measured-Trace Channel

For realism we replay a measured UAV link from DeepSense 6G Scenario 23 [21], [22], in which a drone flies around a ground receiver that records 60 GHz beam powers together with position, altitude and speed. The dataset's power values are uncalibrated, so only their variation relative to the median is used:

$$\gamma_k = \bar{\gamma} + \big(P_k^{\mathrm{meas}} - \mathrm{med}\, P^{\mathrm{meas}}\big) - 10\eta\log_{10}\frac{d_{k,b}}{d_{k,1}}, \tag{3.3}$$

where P_k^{meas} is the recorded beam power in dB on the dataset's arbitrary scale, $\bar{\gamma}$ is an assumed reference SNR, and the last term adjusts the measured link to the distance of

a virtual BS b, with distances in metres so that their ratio is dimensionless. Each UAV replays a different flight segment. Two settings are used: $\bar{\gamma} = 15\,\mathrm{dB}$, a good link, and $\bar{\gamma} = 5\,\mathrm{dB}$, a weak link. They replay identical trajectories, so only the link quality differs.

3.2.3 Packet Erasures

Both channel models draw a separate erasure event for each packet, conditional on the current link state. A single draw determines whether all L parameters in that packet are available to the BS. The erasure probability is

$$p_k = \frac{1}{1 + \exp\big(a(\gamma_k - \gamma_{\mathrm{th}})\big)}, \tag{3.4}$$

with $a = 1$ and $\gamma_{\mathrm{th}} = 3\,\mathrm{dB}$. For the drop policy used in the controlled study, the BS discards the entire upload if its estimated transmission time, computed at the Shannon rate, exceeds 2 s. The BS therefore receives $\tilde{\mathbf{\Delta}}_k = \mathbf{m}_k \odot \mathbf{\Delta}_k$, where the mask $\mathbf{m}_k$ is 1 for delivered coordinates and 0 elsewhere. The BS also records link telemetry for each UAV: SNR, packet delivery ratio (PDR), long-run reliability, distance, speed and handover state.

Choice of the packet-error curve. The base paper [1] models packet errors with the normal approximation for finite-length codes: a k-bit packet sent in n channel uses at linear SNR γ is lost with approximate probability, using the AWGN third-order term $\frac{1}{2}\log_2 n$ [23],

$$p = Q\left(\frac{n\log_2(1+\gamma) - k + \frac{1}{2}\log_2 n}{\sqrt{n\,V(\gamma)}}\right), \qquad V(\gamma) = \big(1 - (1+\gamma)^{-2}\big)\log_2^2 e. \tag{3.5}$$

Our packets carry $L = 1024$ parameters of 32 bits, so $k = 32{,}768$. Choosing the code rate whose capacity threshold is $\gamma_{\mathrm{th}} = 3\,\mathrm{dB}$, $R = \log_2(1 + 10^{0.3}) = 1.58$ bits per channel use ($n \approx 20{,}704$), places the 50% point of (3.5) approximately at 3 dB, the midpoint of our logistic curve. For a single codeword of this length, however, (3.5) is a waterfall: the packet error rate falls from 90% to 10% within 0.11 dB. The SNR replayed from the trace is one sample per round, while the real SNR varies during a round. If that variation is modelled as a Gaussian spread in dB and (3.5) is averaged over it, a spread of

1.7 dB reproduces the logistic curve with $a = 1$ to within 1.0 percentage point of packet error rate over the plotted SNR grid (Fig. 3.2). The logistic curve is thus a numerical approximation to this SNR-averaged normal approximation under the fitted spread. The spread is a fitted modelling assumption, not a measured value, and because the 5 and 15 dB anchors are also assumed, the results are best read against the delivery ratios they produce (Section 4.5). Phase II will use (3.5) directly, with the SNR variation taken from the trace.

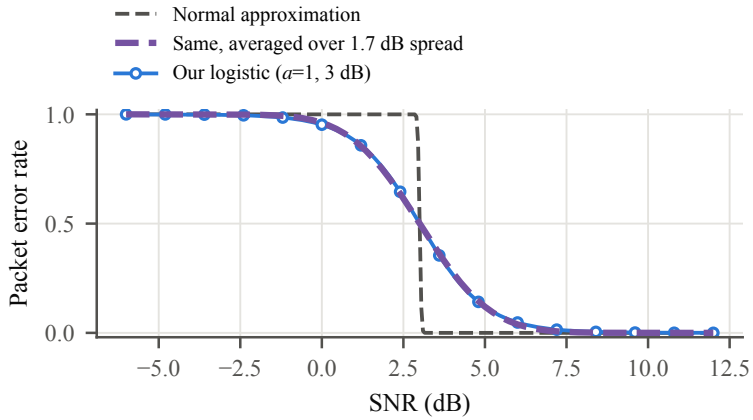


Figure 3.2 Packet error rate against SNR: the normal approximation with the half-log term for a 32,768-bit packet at rate 1.58 bits per channel use, the same curve averaged over a 1.7 dB Gaussian SNR spread, and the logistic curve used in this work.

3.3 Threat Model

A fixed set of $f = 3$ of the $K = 10$ UAVs is Byzantine. They follow the protocol (they train, transmit on time and use the same channel) but replace their update with one of the following (`fl/attacks.py`, `fl/client.py`):

- **Label flip** (data poisoning): train on labels mapped $c \mapsto 9 - c$.
- **Sign flip**: send $-\lambda \boldsymbol{\Delta}_k$ with $\lambda = 5$.
- **ALIE-style** [11]: send $\boldsymbol{\mu} - z\,\boldsymbol{\sigma} \odot \mathrm{sign}(\boldsymbol{\mu})$ with $z = 1.5$, where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are the mean and standard deviation of the honest updates in the attacker's cell.
- **IPM** [12]: send $-\epsilon\boldsymbol{\mu}$ with $\epsilon = 0.5$.

The ALIE and IPM attackers know the honest updates of their cell, which is a strong assumption chosen to stress the defences. The BS's telemetry is assumed authentic; attacks that forge it are left for Phase II.

3.4 Aggregation Rules

All rules receive exactly the same inputs: the received updates, their masks, sample counts and telemetry (`fl/aggregators.py`).

3.4.1 Baselines

FedAvg [2] averages the received updates weighted by n_k, with lost coordinates counted as zeros. **CosAvg**, used only in the preliminary study, is a CosAvg-style baseline inspired by the earlier UAV-FL work of Liao et al. [20]: it weights each update by $\max(0, \cos(\boldsymbol{\Delta}_k, \bar{\boldsymbol{\Delta}}))\, n_k$, where $\bar{\boldsymbol{\Delta}}$ is the FedAvg mean. It is not an exact reproduction of that paper, whose Algorithm 2 sums pairwise cosine similarities and aggregates the top-ranked updates.

Coordinate-wise median [6] and **FLTrust** [7] were defined for complete update vectors; we adapt both to the packet masks. The median takes, for each parameter, the median of the delivered values only (the lower middle value when their number is even). FLTrust computes a root update $\mathbf{r}$ on $\mathcal{D}_0$ with as many local steps as the median UAV in the cell, weights each UAV by $\max(0, \cos)$ to $\mathbf{r}$ on its delivered coordinates, rescales the delivered part to the norm of $\mathbf{r}$ on the same coordinates, and averages each coordinate over the UAVs that delivered it. The root-set design is FLTrust's; the mask handling is our adaptation. FLTrust uses about 78 root optimizer steps per cell-round, matching the median client budget. CCRA and its variants instead train the same root set for one epoch, $\lceil 500/64 \rceil = 8$ steps. The root training budgets are therefore unequal.

Base-paper erasure rules. The base paper [1] models each upload as erased or received and compares two server rules: average only the fresh updates received in the round (its Eq. (4)), or reuse the last local parameters received from a client whose fresh update is missing (its Eq. (5)). We implement both at packet granularity, which lets

them use every delivered packet: *fresh-only* averages each coordinate over the UAVs that delivered it, weighted by n_k; *reuse-old* fills each lost coordinate, and every coordinate of a fully erased upload, with the last parameter value received from that UAV, re-expressed against the current edge model. Neither rule defends against attackers, as the base paper assumes none.

3.4.2 Context-Calibrated Robust Aggregation (CCRA)

CCRA is the aggregator designed in this project. Each BS applies the following steps every round.

1. **Reference update.** Compute $\mathbf{r}$ on $\mathcal{D}_0$, as in FLTrust. Our diagnostics showed that, under non-IID data, the mean or median of the UAV updates is not a reliable reference: attackers and honest UAVs with skewed data had similar cosine similarity to it.

2. **Norm clipping.** Let ρ_k be the RMS of the delivered entries of $\tilde{\mathbf{\Delta}}_k$ and scale the update by $s_k = \min(1,\ \mathrm{med}_j \rho_j/\rho_k)$, with a small constant added to ρ_k to avoid division by zero. For nonzero updates above the median RMS, the scaling caps their delivered RMS at that median.

3. **Masked similarity.** $c_k = \cos\big(\mathcal{S}(s_k\tilde{\mathbf{\Delta}}_k; \mathbf{m}_k),\ \mathcal{S}(\mathbf{r}; \mathbf{m}_k)\big)$, where $\mathcal{S}(\cdot; \mathbf{m})$ selects the same delivered entries from both the client and reference vectors. The score is set to zero when the selected set is empty or either selected vector has zero norm. Its value can still depend on which packets arrived.

4. **Behaviour score.** $b_k = \mathrm{clip}\big(c_k/\mathrm{med}_{j:c_j>0} c_j,\ 0,\ 1\big)$. Only a positive cosine can give a positive behaviour score. If there are no positive cosines in the cell, every score is zero.

5. **Reputation.** With memory $\beta \in [0, 1)$, initial trust 1 and evidence $e_k \in [0, 1]$,

$$T_k \leftarrow T_k + \eta_k(b_k - T_k), \qquad \eta_k = \begin{cases} (1-\beta)\, e_k, & b_k > T_k, \\ 1-\beta, & b_k \le T_k. \end{cases} \tag{3.6}$$

A smaller e_k slows trust recovery when $b_k > T_k$. For a received update with $b_k < T_k$, the same penalty applies regardless of e_k. The rule was designed to prevent poor delivery from softening that penalty.

6. **Gating and averaging.** The gate sets $\omega_k = 0$ below the trust threshold τ. At or above the threshold, ω_k is the product of n_k, T_k and the staleness discount. For coordinate p, the mask selects the eligible weights for the following average. A zero denominator gives a zero update for that coordinate.

$$\mathrm{Agg}_p = \frac{\sum_k \omega_k \, m_{k,p} \, s_k \tilde{\Delta}_{k,p}}{\sum_k \omega_k \, m_{k,p}}. \tag{3.7}$$

The evidence term multiplies five link factors:

$$e_k = \mathrm{PDR}_k \cdot R_k \cdot h_k \cdot \frac{1}{1 + d_k/A} \cdot \frac{1}{1 + |v_k|}, \tag{3.8}$$

Here PDR_k is the current packet delivery ratio and R_k is the long-run reliability. A handover halves the evidence through h_k (otherwise $h_k = 1$). The final two terms reduce the evidence as the UAV–BS distance d_k or the magnitude of the normalized speed v_k increases. The distance is scaled by the configured area side length $A = 800\,\mathrm{m}$.

3.4.3 Ablation Variants

To find out which part of CCRA matters, three variants are compared.

- **CCRA (full)**: all steps, with evidence e_k from telemetry.
- **Mask-only**: $e_k = 1$, but the masks are used in steps 2–6.
- **No-context**: $e_k = 1$ and $\mathbf{m}_k = \mathbf{1}$. This variant treats the zero placeholders for erased coordinates as observed values.

All three variants receive updates subject to the same deadline and staleness policy. The ablation changes only how the aggregator handles erasure masks and telemetry. The code also has an option to correct c_k for the noise expected at the reported SINR. It is

inactive in the controlled study, which adds no residual noise to delivered parameters, so the full-versus-mask-only contrast isolates e_k.

3.4.4 Analysis of the Reputation Update

The asymmetric update (3.6) has a consequence that was not intended. Suppose a UAV's evidence is constant, $e_k = e \in (0, 1]$, its scores $b \in [0, 1]$ are i.i.d. and independent of its current reputation, with $0 < \mathbb{E}[b] < 1$, and $0 \le \beta < 1$. Conditional on the current level T, the expected change in reputation is

$$\mathbb{E}[\Delta T \mid T] = (1 - \beta)\, g_e(T), \qquad g_e(T) = e\,\mathbb{E}\big[(b - T)^+\big] - \mathbb{E}\big[(T - b)^+\big], \tag{3.9}$$

so the drift vanishes where

$$e\,\mathbb{E}\big[(b - T^\star)^+\big] = \mathbb{E}\big[(T^\star - b)^+\big]. \tag{3.10}$$

This balance is an expectile condition [24]: dividing by $1 + e$ gives $q\,\mathbb{E}[(b - T^\star)^+] = (1-q)\,\mathbb{E}[(T^\star - b)^+]$ with $q = e/(1+e)$, the first-order condition that defines the q-expectile of b. Our contribution is to apply this relationship to the reputation rule.

Proposition 3.1. Under the assumptions above, g_e has a unique zero $T^\star(e) \in (0, 1)$, the q-expectile of b with $q = e/(1 + e)$. It equals $\mathbb{E}[b]$ when $e = 1$, is nondecreasing in e, and tends to $\operatorname{ess\,inf} b$ as e decreases to zero.

Proof. g_e is continuous with $g_e(0) = e\,\mathbb{E}[b] > 0$ and $g_e(1) = -\mathbb{E}[1 - b] < 0$, so a zero exists in $(0, 1)$. For $T_2 > T_1$, the change $g_e(T_2) - g_e(T_1)$ lies between $-(T_2 - T_1)$ and $-e(T_2 - T_1)$, so g_e is strictly decreasing and the zero is unique. For $e = 1$, $g_1(T) = \mathbb{E}[b] - T$. Increasing e weakly increases $g_e(T)$ for every T, so it cannot lower the zero. For $T < \operatorname{ess\,inf} b$, the downward term vanishes and $g_e(T) > 0$, so the zero is at least $\operatorname{ess\,inf} b$. For any larger T, the downward term $\mathbb{E}[(T - b)^+]$ is strictly positive while the upward term $e\,\mathbb{E}[(b - T)^+]$ vanishes as $e \to 0$, so the limiting zero cannot exceed $\operatorname{ess\,inf} b$. ∎

$T^\star$ is the equilibrium of the conditional drift, not a statement that the reputation converges almost surely: with a constant step ($\beta = 0.5$) and random scores it keeps fluctuating, and in general $\mathbb{E}[g_e(T)] \neq g_e(\mathbb{E}[T])$. In the simulator the evidence also changes

over time, and scores depend on training, on the other UAVs and on the median normalization, so the constant-evidence, i.i.d. assumptions are an explanatory approximation. Under that approximation, weakening the evidence lowers the equilibrium from the mean score toward the lower end of the score distribution. A finite weak evidence does not put it at the lowest score; that is only the limit as $e \to 0$. An honest UAV with skewed data has noisy scores, so on a weak link this suggests a mechanism by which its trust recovers more slowly and falls below the gate τ more often. An attacker's scores are already near zero, so slowing its rewards adds little. Section 4.5.3 checks whether the measured gate decisions are consistent with this mechanism.

Chapter 4

Experimental Study

This chapter describes how the framework was implemented and verified, the protocol of each experiment, and the results, in the order the work was done: the FedAvg baseline, the preliminary single-seed study, and the controlled 720-run study, followed by a separate 240-run base-paper comparison. Result tables and figures are generated from committed run-level data by the scripts described in Appendix B.

4.1 Implementation and Verification

Each run of the Python/PyTorch simulator saves the configuration dataclass, the per-round history and the final model. Automated tests cover data partitioning, the FedAvg update, attack construction, every robust aggregator, the base paper's two erasure rules, the hierarchical server, handover logic, trace loading and the analysis code. Review of the hierarchical simulator led to three corrections before the controlled runs. A low delivery ratio had dampened both rewards and penalties, so we made the penalty independent of telemetry. We also removed the delivery-ratio input from no-context CCRA and matched the number of FLTrust root optimizer steps to the median local step count in each cell.

4.2 Experimental Protocol

4.2.1 Preliminary Study

The preliminary runs checked the simulator end to end and provided initial estimates of the accuracy differences. At one base station, FedAvg, the CosAvg-style baseline and early CCRA each ran for 30 rounds with the same seed. For each method, we crossed clean training and 30% label flipping with ideal delivery and synthetic jamming.

4.2.2 Controlled Ablation Study

Each of the 10 seeds contributes 72 runs. We test FedAvg, the median, FLTrust and three CCRA variants in every combination of the four attack settings (clean, sign flip, ALIE and IPM) and three channels (ideal, 15 dB trace and 5 dB trace). For each seed we reuse the client partitions, initial weights, attacker identities and trace positions across methods. This gives 720 matched runs on an NVIDIA A100.

Each run lasts 100 rounds with ten UAVs, three base stations and synchronization every five rounds. The client partitions contain 49,500 images in total ($\alpha = 0.5$). A separate 500 images form the trusted root set. The channels operate without a jammer or additive noise on delivered parameters.

Before launching the runs, we specified how to measure the extra effect of telemetry at the weaker trace anchor. For each seed, we compute the full-minus-mask-only accuracy gap at each anchor and average the following difference over sign flip, ALIE and IPM:

$$(\text{CCRA} - \text{mask-only})_{5\,\text{dB}} - (\text{CCRA} - \text{mask-only})_{15\,\text{dB}}.$$

A practically meaningful effect requires the full 95% interval to lie below -1 point or above $+1$ point. We use percentile intervals from 10,000 bootstrap resamples of the paired seeds, the unit of replication. Numerical divergence contributes 10% accuracy, the chance level for the ten-class task. The remaining contrasts are exploratory and use unadjusted intervals.

4.2.3 Follow-up on the Base Paper's Erasure Rules

For the follow-up we substitute fresh-only and reuse-old (Section 3.4.1) for the six ablation methods. The fixed comparison retains the original seeds, attacker identities, three channel conditions and four attack settings. The two rules contribute 120 runs each, for a validated total of 240 outside the prespecified ablation. The follow-up seeds were used for evaluation without parameter tuning. Both rules use the same 49,500 client images and reserve the same disjoint 500 root images, although neither uses a trusted reference update.

The follow-up was executed through SwarmCompute on Windows NVIDIA workers. Final attempts used an RTX 2060 for 238 runs and a GTX 1650 for two; 57 final attempts resumed from centrally validated checkpoints. Configurations, source and dataset fingerprints, terminal records and all 120 within-study scientific pairs were checked. Runtime profiles and artifact checksums are retained with the compact results. Hardware and software versions differ from the A100 ablation, so cross-study rankings are descriptive. The paired fresh-only versus reuse-old intervals resample the ten seed pairs within this follow-up; they do not make the two studies a single prespecified experiment.

4.3 FedAvg Baseline Results

The first objective was a working FL pipeline without any defence. Ten UAVs trained TacticalEdgeCNN for 50 rounds with two local epochs, SGD (learning rate 0.01, momentum 0.9, batch size 64), on one BS with an ideal channel. Table 4.1 and Fig. 4.1 show the result.

Table 4.1 FedAvg baseline accuracy after 50 rounds

Data partition	Final test accuracy (%)	Best test accuracy (%)
IID (5,000 balanced images per UAV)	62.6	62.7
Non-IID, Dirichlet $\alpha = 0.5$	60.2	60.6

Non-IID data cost about 2.4 percentage points, and neither curve had fully levelled off by round 50. These accuracies are not direct targets for the later experiments, which use 30 or 100 rounds, one local epoch and, in the controlled study, a hierarchical system;

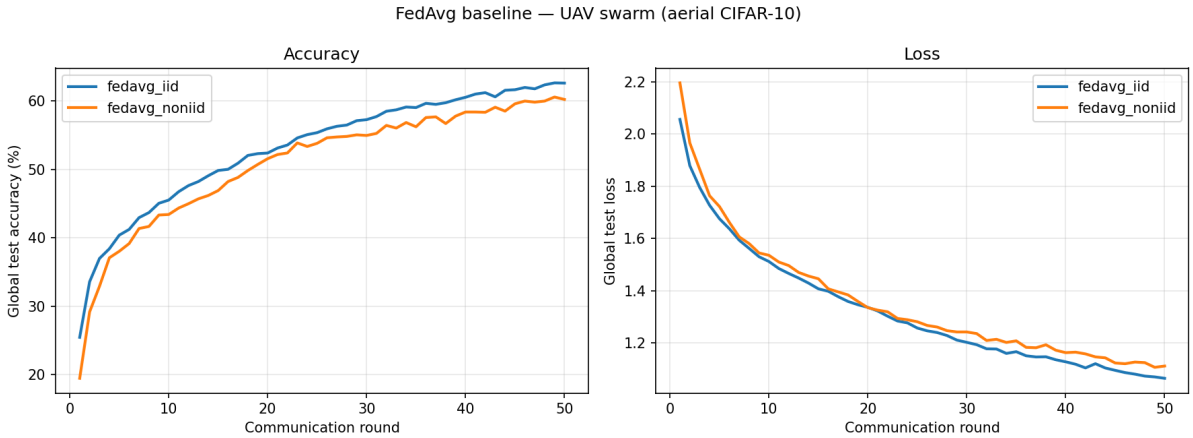


Figure 4.1 Test accuracy and loss of the FedAvg baseline over 50 rounds.

robust methods are instead compared with matched clean controls within each experiment.

4.4 Preliminary Study Results

The preliminary study compared FedAvg, the CosAvg-style baseline (inspired by Liao et al. [20], not an exact reproduction) and an early version of CCRA with one BS, 30 rounds, one local epoch and a single seed. Three of the ten UAVs flipped their labels, and in the jammed scenarios a ground jammer targeted three honest UAVs. Table 4.2 lists the final accuracy (mean of the last five rounds), and Figs. 4.2 and 4.3 show the accuracy and the weight each method gave to the attackers.

Table 4.2 Preliminary study: final accuracy (%), single seed, 30 rounds. CosAvg is our CosAvg-style baseline, not the published method.

Scenario	FedAvg	CosAvg	CCRA (early)
No attack, ideal channel	46.5	**47.1**	44.5
No attack, jammed lossy channel	44.6	**45.6**	43.1
30% label flip, ideal channel	37.6	36.6	**43.3**
30% label flip, jammed lossy channel	36.8	39.7	**43.0**

The early CCRA kept about 43% accuracy in both attacked scenarios. Averaged over the last ten rounds, attackers received 1.9% of its aggregation weight on the ideal channel (CosAvg-style: 19.1%; FedAvg: 26.0%) and 8.4% on the jammed lossy channel (CosAvg-style: 21.5%; FedAvg: 27.5%). With no attack, early CCRA was 2.0 points below FedAvg on the ideal channel and 1.5 points below it on the jammed channel; the

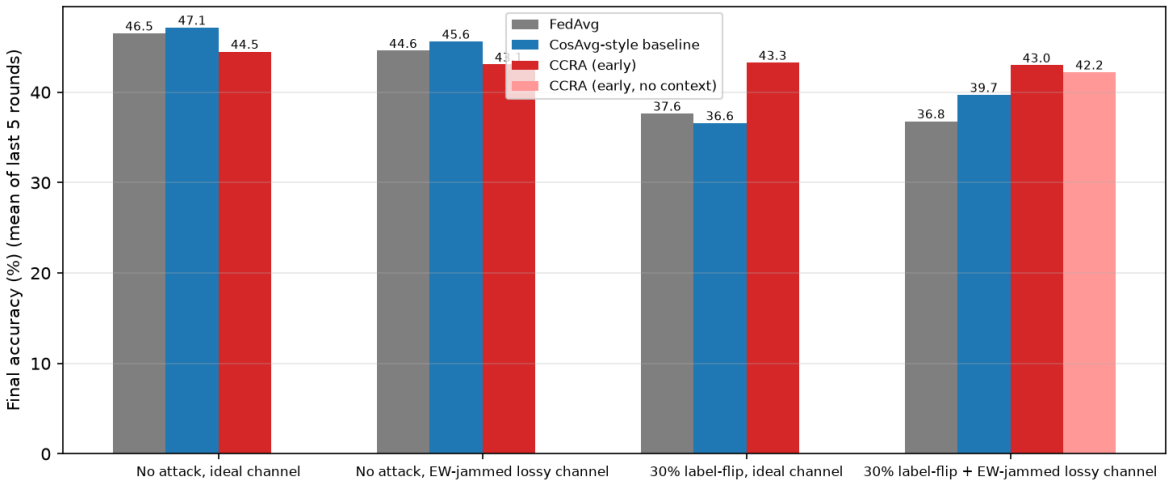


Figure 4.2 Preliminary study: final accuracy in the four scenarios.

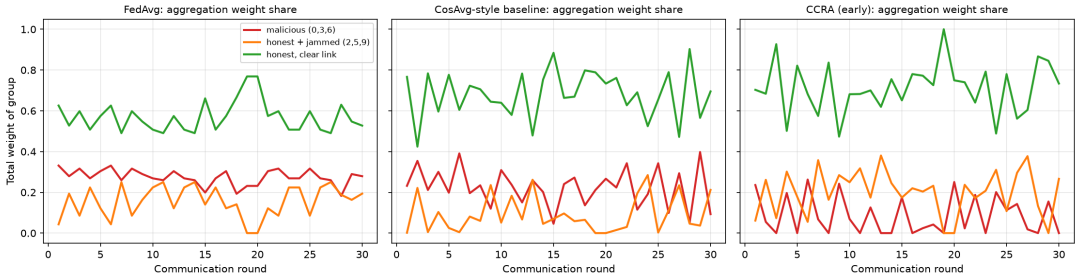


Figure 4.3 Preliminary study: aggregation weight given to attackers and to jammed honest UAVs under label flipping on the jammed channel.

corresponding costs relative to the CosAvg-style baseline were 2.7 and 2.5 points. These results were encouraging, but they could not support a claim:

- they came from one seed, so the size of the random variation was unknown;
- the channel-aware part of CCRA added only 0.8 points over its no-context ablation (42.2 to 43.0%), and with one seed the uncertainty in that difference could not be estimated;
- CCRA used a trusted root set while CosAvg did not, so the comparison mixed two differences;
- the reputation rule was revised when the system was extended to hierarchical FL, so the early CCRA is not the exact version evaluated in the controlled study.

The paired ablation isolates the effects of masking and telemetry.

4.5 Controlled Ablation Study Results

Validation accounted for all 720 planned outcomes, including numerical divergences. The trace delivered on average 99.8% of packets at 15 dB and 65.0% at 5 dB. The two trace anchors share a geometry with neighbouring base stations roughly 14 m from the central station. The ideal condition instead uses random walks and neighbours 240 m from the centre. The change in conditions also changes association behaviour: the ideal runs average 43.5 handovers, compared with 0.7 per trace run. Comparisons with the ideal setting therefore combine packet delivery, geometry and mobility effects.

4.5.1 Stability and Overall Accuracy

Table 4.3 pools all conditions to show divergence counts and overall accuracy. Figure 4.4 restricts the accuracy comparison to attacked runs and groups them by channel.

Table 4.3 Outcomes over all 720 runs (120 per method). Diverged runs count as 10%.

Method	Diverged	All runs (%)	Attacked, 5 dB (%)
FedAvg	41/120	37.3	29.4
Median	26/120	29.3	21.3
FLTrust	0/120	50.6	**50.1**
No-context	0/120	51.4	45.3
Mask-only	0/120	**52.8**	49.5
CCRA (full)	0/120	51.1	46.0

FedAvg diverged in 41 and the coordinate-wise median in 26 of their 120 runs; FedAvg diverged in all 30 sign-flip runs. None of the four methods that use a trusted reference update (FLTrust and the three CCRA variants) diverged in its 120 runs. A plausible explanation for the median's failures, not tested by a separate diagnostic, is that each cell holds only a few UAVs, so one attacker in a cell is already a large fraction of the values being compared. Over all runs the mask-only variant has the highest accuracy, 52.8%, against 37.3% for FedAvg. Table 4.4 breaks the result down by attack.

Two points in Table 4.4 deserve attention. First, with no attack, plain FedAvg is the most accurate method on the ideal and 15 dB channels; every robust method pays a cost of roughly two to eight points for discarding or down-weighting honest updates.

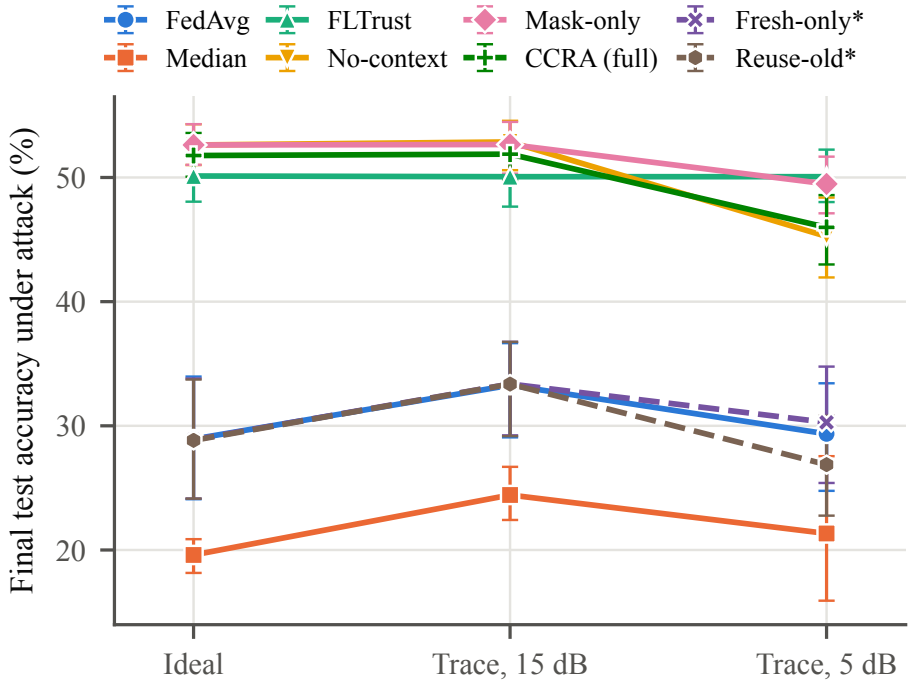


Figure 4.4 Final accuracy under attack (mean over sign flip, ALIE and IPM and 10 seeds). Error bars: 95% seed-bootstrap intervals. Dashed, starred rules are from the separate 240-run follow-up; cross-study rankings are descriptive.

Second, on the 5 dB channel the masked variant is slightly ahead of FedAvg even without an attack, consistent with FedAvg's zero-filled lost coordinates now hurting it.

4.5.2 Which Kind of Channel Awareness Helps

Figure 4.5 shows the paired contrasts that separate the two channel mechanisms.

Erasure masking. On the ideal channel nothing is lost, so mask-only and no-context are identical. As the link weakens the masks matter more, and at 5 dB masking adds 4.2 points over the no-context baseline, which treats lost packets as zeros (95% CI [2.2, 5.9]). Treating lost coordinates as zeros shrinks honest updates, distorts their similarity scores and pulls the average towards zero; ignoring them avoids all three effects.

Telemetry in the reputation. The prespecified primary estimate, the telemetry effect at 5 dB minus that at 15 dB, is −2.7 points ([−3.9, −1.5]). Even the upper bound falls below −1 point, meeting the prespecified criterion for an additional accuracy penalty at the weaker anchor. At 5 dB, full CCRA scored 46.0% and mask-only scored 49.5%, a gap of −3.5 points ([−5.1, −2.1]). In the attacked ideal runs, where delivery is complete,

Table 4.4 Final test accuracy averaged over seeds (%). Within each attack setting, bold marks the highest mean for each channel.

Attack	Method	Ideal	Trace, 15 dB	Trace, 5 dB
No attack	FedAvg	**59.2**	**59.3**	54.1
	Median	52.5	52.7	49.7
	FLTrust	52.5	51.5	52.9
	No-context	57.1	56.7	51.1
	Mask-only	57.1	57.5	**54.7**
	CCRA (full)	55.4	56.6	52.6
Sign flip	FedAvg	10.0	10.0	10.0
	Median	10.0	19.6	12.4
	FLTrust	50.0	49.9	**50.6**
	No-context	**54.2**	**53.8**	46.1
	Mask-only	54.2	53.6	49.9
	CCRA (full)	52.5	52.8	46.5
ALIE	FedAvg	49.8	46.9	42.0
	Median	35.4	28.4	28.1
	FLTrust	49.9	50.6	49.6
	No-context	**52.9**	52.6	46.0
	Mask-only	52.9	**52.7**	**50.8**
	CCRA (full)	52.8	51.9	46.9
IPM	FedAvg	27.0	42.9	36.0
	Median	13.4	25.3	23.5
	FLTrust	50.5	49.7	**50.0**
	No-context	**50.7**	**52.3**	43.6
	Mask-only	50.7	51.7	47.7
	CCRA (full)	50.0	51.0	44.5

the gap is −0.8 points.

Both together. Full CCRA against no-context combines a gain and a loss; at 5 dB the net effect is small and its interval includes the equivalence region. The earlier impression that "channel context helps" came from this combination, which hid two opposite effects.

4.5.3 Mechanism: Who the Gate Rejects

Section 3.4.4 predicted that slowing rewards on weak links would push honest UAVs' reputations down without doing much to attackers. Figure 4.6 is consistent with that pattern.

With telemetry, the gate rejected 41.2% of honest updates against 13.8% without it,

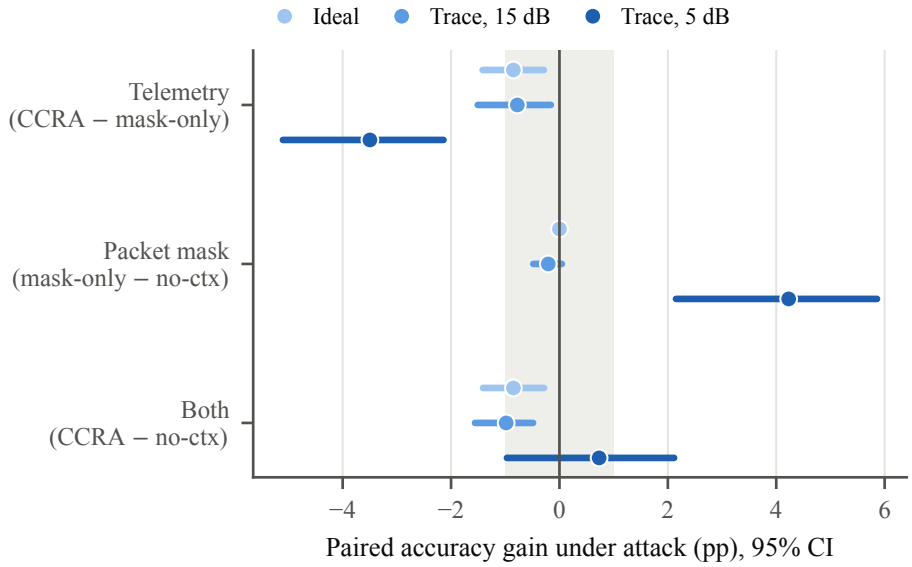


Figure 4.5 Paired accuracy differences under attack with 95% bootstrap intervals. The grey band is the prespecified ±1 point region of practical equivalence.

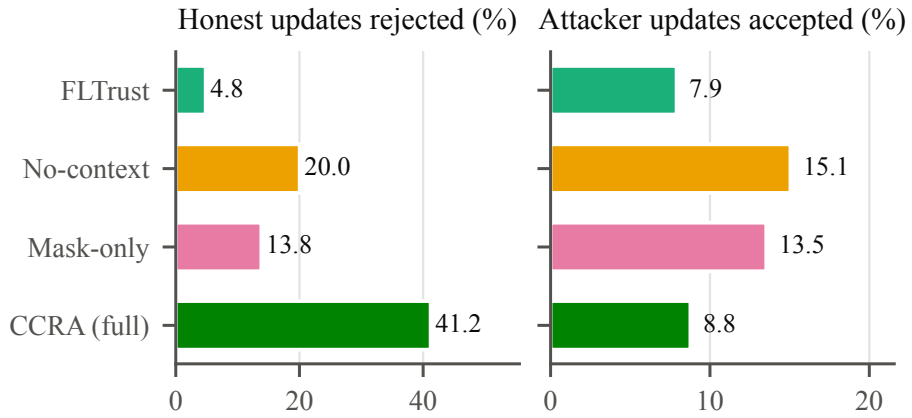


Figure 4.6 Gate decisions under attack at 5 dB, as a percentage of received updates, averaged over attacks and seeds.

while the share of attacker updates accepted fell only from 13.5% to 8.8%. The accuracy loss from telemetry was accompanied by this rise in honest-update rejection. The rule gave up a large amount of honest data to filter slightly more attacker data, and it did so precisely when honest data were already scarce because of packet loss. These rates are descriptive. They agree with the analysis in Section 3.4.4, but they do not by themselves prove that it is the cause.

4.5.4 Comparison with FLTrust

At the 5 dB anchor under attack, FLTrust exceeds full CCRA by 4.1 points (50.1% versus 46.0%; Table 4.3). FLTrust is the closest established method, since it uses the same root set, although with more root training steps. Its gate rejected only 4.8% of honest updates at 5 dB. It has no reputation memory, since each round is judged on its own, but its norm matching, score normalization, weights and gating also differ from CCRA's, so the lower rejection cannot be attributed to memory alone. In an exploratory paired comparison, mask-only was 2.6 points better than FLTrust under attack at 15 dB ([0.9, 4.5]), while the 5 dB comparison was inconclusive (-0.6 points, $[-2.0, 1.0]$): the interval supports neither a difference nor practical equivalence, since it is not entirely inside ± 1 point. The different root training budgets and aggregation rules also limit what this comparison can establish about reputation memory.

4.6 Completed Base-Paper Comparison

The 240 terminal outcomes comprise 157 successful runs and 83 numerical divergences. Fresh-only diverged in 41 of 120 runs, and reuse-old in 42 of 120. All 60 sign-flip runs diverged; the remaining 23 divergences occurred under IPM. No clean or ALIE run diverged. These outcomes are retained at 10% final accuracy, rather than excluded or retried into successful runs. All-run accuracy is 37.7% for fresh-only and 36.4% for reuse-old.

Table 4.5 shows identical mean accuracy on the clean ideal channel, where there are no missing packets to reuse. On clean stressed links, fresh-only attains 56.9% versus 51.5% for reuse-old: a gain of 5.4 percentage points (95% CI [4.1, 6.7]). Under attack at 5 dB, the chance-scored means are 30.3% and 26.9%, respectively, a gain of 3.4 points ([0.9, 6.6]). The interval describes a mean over the three attacks within each seed, not 30 independent replicates.

Retaining old parameters does not improve this finite-horizon stressed-channel experiment, and neither rule provides Byzantine robustness. This is compatible with the scope of Shirvanimoghaddam et al. [1]: its analysis assumes honest clients and whole-

Table 4.5 Separate base-paper follow-up: ten paired seeds per cell. Final accuracy includes numerical divergences at chance (10%). Intervals are exploratory paired seed-bootstrap intervals, with no multiplicity adjustment.

Attack	Channel	Fresh (%)	Reuse (%)	Fresh − reuse: pp [95% CI]
No attack	Ideal	59.0	59.0	+0.0 [+0.0, +0.0]
	15 dB	59.4	59.4	+0.0 [−0.2, +0.3]
	5 dB	56.9	51.5	+5.4 [+4.1, +6.7]
Sign flip	Ideal	10.0	10.0	+0.0 [+0.0, +0.0]
	15 dB	10.0	10.0	+0.0 [+0.0, +0.0]
	5 dB	10.0	10.0	+0.0 [+0.0, +0.0]
ALIE	Ideal	49.7	49.5	+0.2 [+0.0, +0.3]
	15 dB	47.2	47.2	+0.0 [−0.3, +0.3]
	5 dB	43.3	39.8	+3.5 [+0.5, +6.5]
IPM	Ideal	26.9	26.9	+0.0 [−0.1, +0.0]
	15 dB	42.9	42.9	+0.0 [−0.1, +0.2]
	5 dB	37.5	30.8	+6.7 [+1.4, +15.0]

upload erasures, whereas the present implementation uses packet erasures in a hierarchical non-IID CNN task with adversarial updates. These results are an adaptation study, not an exact reproduction or a refutation of the original convergence theorem.

Figure 4.4 now includes both rules alongside the original six methods. The original ablation's findings about masking and telemetry are unchanged.

4.6.1 Our Method against the Base Paper

Both studies use the same seeds, data partitions and attacker identities, so mask-only CCRA can be compared with the base paper's better rule, fresh-only, seed by seed. Under attack at 5 dB, mask-only attains 49.5% against 30.3% for fresh-only, a paired gain of 19.2 percentage points (95% CI [15.9, 23.0]). Without attack at 5 dB, fresh-only is ahead, 56.9% against 54.7%, a signed difference of −2.1 points ([−2.9, −1.4]); the signed contrast is mask-only minus fresh-only. FLTrust, which also uses a trusted reference, scored 19.8 points above fresh-only in the attacked 5 dB comparison. Fresh-only already restricts its average to delivered coordinates. Both contrasts also include differences in scoring and weighting. The controlled comparison with zero-filled no-context CCRA estimates the effect of masking separately.

The two studies ran on different hardware (an A100 against RTX 2060 and GTX 1650 workers) and software versions, so these contrasts are cross-study and exploratory. Complete delivery makes fresh-only equivalent to FedAvg, giving us a cross-check on the ideal runs. The same runs diverged in both studies. For the 24 remaining ideal pairs, the absolute accuracy difference averaged 0.3 points and reached at most 1.1 points. This check measures agreement only under complete delivery. The hardware contribution to the 5 dB comparison remains unmeasured.

The coordinator recovered interrupted jobs separately from the divergence analysis. It verified the checkpoints before requeuing clean stops recorded as application failures. Windows progress-file permission errors triggered bounded retries with the same experiment settings. The audit history retains the failed attempts, and validation accepted exactly one terminal record for each planned job. A resumed job restores the saved training and random-generator states. Bitwise agreement between GPUs was outside the scope of that validation.

4.7 Discussion

4.7.1 What the Results Show

1. FLTrust and the three CCRA variants completed all 480 runs without numerical divergence. The root set is one of several design choices that distinguish these methods from FedAvg and the median.

2. In the attacked trace runs, masking adds little at 15 dB, where nearly every packet arrives. At 5 dB, it raises accuracy by 4.2 points. The study measured accuracy and stability. The computational and communication costs of masking remain unmeasured.

3. At 5 dB, telemetry slowed trust recovery while the penalty for a low score stayed the same. The drift calculation in Section 3.4.4 predicts that this asymmetry lowers equilibrium trust as evidence weakens.

4. In the erasure-rule follow-up, fresh-only gave higher accuracy at 5 dB than reuse-

old. Every sign-flip run diverged with either rule, even with packet-level erasure handling.

4.7.2 Why a Negative Result Is Useful

The preliminary study suggested that CCRA's channel context was its key ingredient. Had that claim been carried forward, Phase II would have been built on a component that harms accuracy. The prespecified design, paired seeds and chance scoring of diverged runs made it possible to reject the claim with confidence, and the analysis in Section 3.4.4 points to the specific rule that must change.

4.7.3 Threats to Validity

- **Channel realism.** The trace comes from one measured BS; the other BSs are virtual, and the 5 and 15 dB anchors are assumptions, not calibrated values. The virtual neighbours are about 14 m from the reference BS (about 28 m from each other). In the separate 60-round channel validation, the median SNR at the nominal 5 dB anchor is 8.6 dB, not 5 dB. There are no retransmissions. The trace replays average 0.7 handovers. Comparisons with the ideal condition also reflect the different geometry and association patterns. The effect of frequent handovers in trace replay remains an open question.
- **Scope of learning task.** The experiment uses one CNN trained on transformed CIFAR-10 images as a proxy for aerial imagery. The swarm size and attack strengths are fixed.
- **Trust assumptions.** The simulation gives each BS access to a root set assumed to be clean. It leaves trusted execution and attestation to a future implementation.
- **Attacker model.** The comparison assumes authentic telemetry and fixed poisoning strategies. An adaptive attacker that targets trust over multiple rounds, as studied by Liu et al. [13], is outside this design.
- **Statistics.** Intervals reflect ten paired seeds. Secondary contrasts use unadjusted intervals. Phase II confirmation requires fresh seeds.

- **Execution environment.** The A100 ablation and Windows-worker follow-up used different GPU/software profiles and source fingerprints. Checkpoint recovery was verified, but hardware-controlled cross-study effects and runtime speedups were not measured.

Chapter 5

Conclusions and Future Scope of Study

5.1 Summary of the Work

This project built a simulation framework for federated learning over a UAV swarm whose updates travel over lossy air-to-ground links and in which some UAVs are Byzantine. The framework supports hierarchical aggregation at three base stations, handover, packet erasures from a synthetic jammed channel or a measured UAV trace, five attacks and twelve selectable aggregation rules (FedAvg, the CosAvg-style baseline, median, trimmed mean, Krum, Multi-Krum, FLTrust, three CCRA variants and the base paper's two erasure rules), with automated tests. The controlled study evaluated six methods; a separate 240-run follow-up evaluated the two base-paper rules. A FedAvg baseline was established first. A reputation-based aggregator, CCRA, was then designed with two forms of channel awareness, erasure masking and telemetry-weighted reputation. After a promising but inconclusive single-seed study, a prespecified, paired study of 720 runs separated the two forms and measured each one's effect. The results have been written up as a manuscript in IEEE letter format.

5.2 Conclusions

1. Non-IID data reduced FedAvg accuracy from 62.6% to 60.2% in the baseline, and a 30% label-flip attack reduced it by 8.9 points on the ideal channel and 7.8 points on the jammed channel in the preliminary study (last-five-round means).

2. None of the four aggregators that judge updates against a trusted reference update diverged in its 120 runs (480 runs combined), whereas FedAvg and the coordinate-wise median diverged in 41 and 26 of 120 runs each.

3. Ignoring lost coordinates instead of treating them as zeros (erasure masking) improved accuracy under attack on a 5 dB link by 4.2 percentage points (95% CI [2.2, 5.9]).

4. Using link telemetry to slow reputation rewards had a meaningful negative effect: the prespecified primary estimate was -2.7 points ($[-3.9, -1.5]$), accompanied by a rise in honest-update rejection from 13.8% to 41.2%.

5. A plausible cause is structural. With slowed rewards but full-rate penalties, the constant-evidence drift analysis (3.10) shows that weaker evidence lowers the equilibrium of an honest UAV's reputation from its mean score toward the lower tail of its scores, and the measured gate decisions are consistent with this mechanism.

6. In the exploratory comparison under attack, mask-only was 2.6 points ahead of FLTrust at 15 dB ([0.9, 4.5]). At 5 dB, the interval was too wide to distinguish an accuracy difference from practical equivalence.

7. Against the base paper, mask-only CCRA was 19.2 points more accurate than its better erasure rule under attack at 5 dB ([15.9, 23.0]), with a signed clean contrast of -2.1 points (mask-only minus fresh-only; $[-2.9, -1.4]$). The corresponding FLTrust gain over fresh-only under attack was 19.8 points. These cross-study comparisons combine several design differences, including the use of a trusted reference.

8. On clean 5 dB runs, fresh-only scored 5.4 points higher than reuse-old ([4.1, 6.7]). Fresh-only and reuse-old diverged in 41 and 42 of 120 runs respectively, including all sign-flip runs. The base paper's convergence analysis concerns honest clients [1].

5.3 Scope for Future Work (Phase II)

1. **Replicate the erasure-rule comparison.** Run both sets of methods on the same hardware and software with new seeds. Compare masking with reusing a UAV's last

delivered parameters subject to a trust check, and measure whether reuse retains poisoned values.

2. **Revise the reputation update.** Compare symmetric weighting of rewards and penalties with scoring against each UAV's recent history. Test whether either change preserves trust for honest UAVs with low packet delivery.

3. **Hold out the confirmation runs.** Freeze the revised rule before measuring its performance on new seeds and channel settings.

4. **Test frequent handovers.** Extend the topology to more base stations and use trajectories that trigger frequent handovers. Measure whether telemetry changes accuracy and honest-update rejection in those runs.

5. **Test the trust assumptions.** In simulation, allow adversaries to vary their behaviour across rounds or supply false link telemetry. Measure the resulting gate decisions and final accuracy.

6. **Network-level simulation.** Replace the packet-error model with a network simulator such as NS-3 to capture medium access, retransmissions and latency.

7. **Generalization.** Repeat the key comparisons with another dataset, a larger model and a larger swarm, and submit the manuscript.

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Experimental Settings

Table A.1 Settings of the controlled study

Parameter	Value
UAVs K, base stations B	10, 3
Byzantine UAVs f	3, fixed per seed across all methods
Rounds; synchronization period E	100; 5
Local training	1 epoch, SGD, learning rate 0.01, momentum 0.9, batch 64, weight decay 5×10^{-4}
Model	TacticalEdgeCNN, 156,074 trainable parameters, no normalization layers
Data	Aerial-transformed CIFAR-10, Dirichlet $\alpha = 0.5$, 49,500 training images to UAVs
Root set $\mathcal{D}_0$	500 images (50 per class), disjoint from all UAV data
Reputation	$\beta = 0.5$, gate $\tau = 0.3$, initial trust 1
Staleness discount	$\exp(-0.5 \cdot \text{staleness})$; late updates dropped
Packets	$L = 1024$ parameters, 32 bits per parameter
Packet errors	logistic, slope $a = 1$, threshold $\gamma_{\text{th}} = 3\,\text{dB}$
Round deadline; bandwidth	2 s; 20 MHz
Handover	3 dB hysteresis, 2-round time to trigger
Trace	DeepSense 6G Scenario 23, stride 5 samples per round, segments ≥ 200 samples
Reference SNR $\bar{\gamma}$	assumed anchors: 15 dB (good link), 5 dB (weak link)
Jammer; parameter noise	none; none
Attacks	sign flip $\lambda = 5$; ALIE-style $z = 1.5$; IPM $\epsilon = 0.5$
Seeds	41–50

Reproducing the Results

All code, the run-level summary results and this report are in the repository https://github.com/sasank-555/uav-swarm-fl. The raw DeepSense 6G data are access-controlled and must be obtained from the dataset provider; per-run model checkpoints are kept locally and are not distributed. The training-source commit pinned for the controlled study is recorded in `results/CONTEXT_STUDY_FINAL.md`.

Table B.1 Repository layout

Path	Contents
`fl/`	Simulator: data, model, client, channel, topology, attacks, aggregators, server
`tests/`	Automated simulator, coordinator and analysis tests
`run_context_study.py`	Plans and runs the 720-run controlled study
`analyze_context_study.py`	Prespecified analysis with paired bootstrap
`results/context_study/`	Committed summary CSV and analysis of the controlled study
`results/context_final/`	Local output of the final-profile reproduction command
`results/base_paper_comparison/`	Compact evidence, paired estimates and validation of the 240-run follow-up
`scripts/make_paper_figures.py`	Builds every figure, table and number in the paper and this report
`paper/`	Manuscript in IEEE letter format
`report/`	This report

Run these commands from the repository root. The final profile writes to `results/context_final/`. The published compact summaries are in `results/context_study/`.

```
pip install -r requirements.txt
python prepare_data.py
python scripts/prepare_deepsense23.py /path/to/Scenario23 --out
    data/channel_traces/deepsense23.npz
```

```
python run_context_study.py --profile final --device cuda --parallel 4 --out-dir
    results/context_final
python analyze_context_study.py results/context_final
```

To rebuild figures and numbers from the committed summaries, then compile the letter and report, use:

```
python scripts/make_paper_figures.py
(cd paper && tectonic -X compile main.tex)
(cd report && tectonic -X compile main.tex)
```

These last three commands require no new training. To plot a fresh reproduction instead, pass `--csv results/context_final/context_runs.csv` to the figure script; its default is the committed `results/context_study/context_runs.csv`.

The completed base-paper follow-up uses frozen training revision `ad06055` and the same runner with `--aggs bp_fresh bp_reuse`. Its full checkpoints, coordinator audit and prepared inputs remain local. Compact publication checks source/data fingerprints, terminal histories and all 120 paired client-partition summaries. Recompute the follow-up intervals from the committed summary data without training:

```
python scripts/analyze_base_paper.py --csv
    results/base_paper_comparison/base_paper_runs.csv
```

Then repeat the figure-generation and document-build commands above. The result CSV records the final attempt's runtime profile and resumed round; an attempt that resumed may include earlier work from a different device. The source fingerprints of both studies are retained separately.